

SECTION 4 – SITE ENVIRONMENT

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4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS

4-2-1 System Suite

Note

If specifications in this section are not strictly adhered to, failures of the ACGD modules in the ACGD/PDU Cabinet may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmentally controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8°F (21°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS FOR SYSTEM WITH ACGD/PDU CABINET

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Equipment Room at Inlet to Equipment	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)**
Magnet Room	59–69.8 (15–21)	5 (3)	30–60*	5	5 (3)
Operator's Control Room	59–89.6* (15–32)*	5 (3)	30–75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

4-2-2 Outdoor TSCC & TGWC Operating Environment

The Outdoor TSCC and Outdoor TGWC are designed to be located external to the building and operate in environments meeting the following specifications.

- Operating Ambient Temperature: –22°F (–30°C) to 122°F (50°C)
- Operating Humidity: 5–100%

4-3 AIR COOLING REQUIREMENTS

Table 4-2 contains the heat output of the upgrade equipment listed in the typical site location. These values do not include people, lights, magnet equipment cabinets, and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Physical placement of the air conditioning equipment (compressor, etc.) is an important factor due to the homogeneous field requirements of the magnet. Therefore, it is important this equipment be located outside the 10 gauss line. Refer to Section 3, MAGNETIC FIELD CONSIDERATIONS, for plot of gauss lines.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal, location of coil/bore cooling fans intake.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch access flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	MAGNET ROOM (SEE NOTE 1)		EQUIPMENT ROOM (SEE NOTE 2)		OPERATOR/CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	8189	2400				
Patient Blower Box	1366	400				
Main Disconnect			900	264		
SRF Cabinet			8530	2500		
ACGD/PDU Cabinet			34,120	10,000		
System Cabinet			6140	1800		
Twin Accessory Cabinet			2354	690		
TSCC Configurations for Magnet & Body Coil water cooling:						
• Outdoor Air Cooled Remote Control Panel (See Note 3 & 5)			34	10		
• Water Cooled (See Note 3 & 6)			12,000	3517		
• Indoor Air Cooled (See Note 3)			120,000	35,140		
TGWC Configurations for only Body Coil water cooling (requires site cooling for Shield Cooler Compressor):						
• Outdoor Air Cooled Remote Control Panel (See Note 4 & 5)			34	10		
• Water Cooled TGWC (See Note 4 & 6) RCP for water cooled TGWC			10,240 34	3000 10		
• Indoor Air Cooled (See Note 4)			61,000	17,880		
Shield/Cryo Cooler Compressor			Heat dissipation to air negligible			
Magnet Monitor			205	60		
Operator Workspace (See Note 7) with LCD Color Monitor					4095	1200
Magnet Monitor UPS* & Modem			450	132		
*MNS Cabinet			2730	800		
*Advantage Workstation Option					See Note 8	

Note * Optional equipment.

† Maximum heat output is defined for temperature and humidity as defined in Table 4-1.

1 Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room specifications as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contain a patient cooling fan.

2 FOR EQUIPMENT ROOM ONLY: The air cooling load averaged over a working day (~12 hours) is typically 1/2 of the maximum value.

3 The TSCC provides water cooling for the Shield/Cryo Cooler Compressor Cabinet (MS5) and the Gradient Coil.

4 The TGWC provides water cooling for the Gradient Coil only. **When a TGWC is selected then the customer/site must still provide water cooling for the Shield/Cryo Cooler Compressor Cabinet.**

5 The Outdoor Air Cooled TSCC and Outdoor Air Cooled TGWC are designed to be installed external to building and exhaust heat to outside air. A Remote Control Box is located in the Equipment Room.

6 The water cooled TSCC or TGWC require customer provided chilled water cooling in addition to the air cooling, refer to Section 4-4 Water Cooling Requirements for details.

7 Operator Workspace equipment includes the following: LCD Color Monitor, Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table

8 Refer to Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation.

4-4 WATER COOLING REQUIREMENTS

4-4-1 Gradient Coil Temporary Backup Water Cooling Recommendation

There are no options available to support temporary backup water cooling for the TRM Gradient Coil. Water cooling must be provided by the cooling cabinet (TSCC or TGWC) supplied with the system to prevent contamination/damage to the coil and for proper image quality.

4-4-2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet (MS5). The backup cooling design can utilize open loop city water only as temporary backup during loss of the closed loop water cooling from the TSCC or customer provided water cooling to the Shield/Cryo Cooler Compressor. Long term open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching that can take place which will eventually contribute to failure. Water system capacity must be selected to make sure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Cooler Compressor Cabinet, refer to Tables 4-3 and 4-4 and Illustration 4-1 for water cooling specifications.



Switching the Shield/Cryo Cooler Compressor inlet/outlet cooling from the TSCC to a temporary water backup supply will result in approximately 1.5 gallons (5.5 liters) of 50% mixture of Dowfrost HD and de-ionized water being discharged. This discharge may have site impacts due to local regulatory codes. Make sure to understand and follow local regulatory requirements when designing and implementing a temporary backup water system.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

Note

Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

TABLE 4-3
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.

4-4-2 Shield/Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation (Continued)

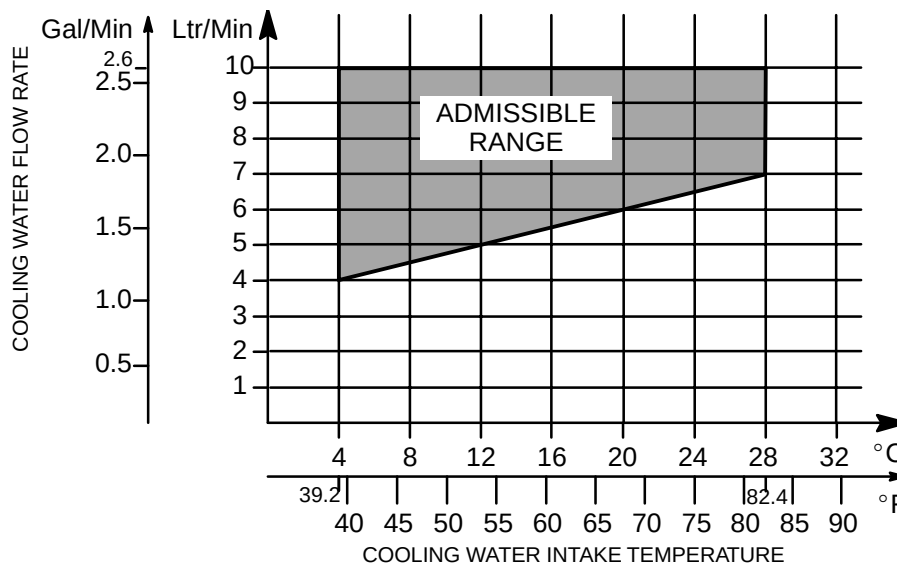
TABLE 4-4
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Note 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28)	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2 & 3	7.5 (0.5) [52] at minimum flow rate See Note 5	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2 & 3	47 (3.2) [324] at maximum flow rate See Note 5	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The backup cooling system design must have allowances for pressure/temperature changes due to distance located from the Shield/Cryo Cooler Compressor.

- 1 Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- 2 Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- 3 Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- 4 Water flow rate and temperature rise value are based on water.
- 5 Pressure drop values based on new system, may rise due to calcification.
- 6 Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.



SHIELD/CRYO COOLER COMPRESSOR REQUIREMENTS FOR TEMPORARY BACKUP WATER COOLING
ILLUSTRATION 4-1

4-4-3 Water Cooled TSCC Requirements

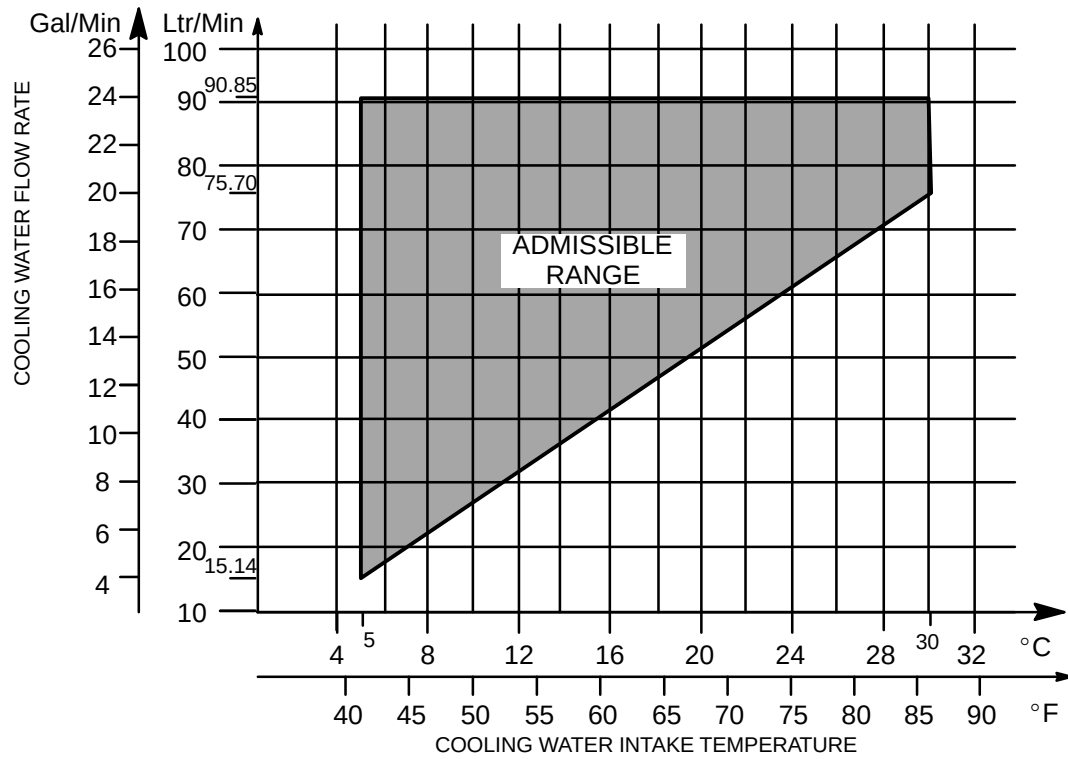
The water cooled configuration of the Twin System Cooling Cabinet (TSCC) contains a dedicated, closed loop, liquid-to-liquid water chiller system providing water cooling for the Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil. The water cooled TSCC configuration requires customer provided water cooling, refer to Table 4-5 and Illustration 4-2 for cooling water requirements. The water cooled TSCC has 1-1/2 inch (38 mm) NPT female connections at the lower back of the TSCC unit for customer water cooling connection. Refer to Illustration 2-13 for water supply and return connection location details. The water supply and return lines must be cleared of debris (solder, dirt, etc.) before connecting to the TSCC, refer to **APPENDIX C Prerequisites for Scheduled Start-up of Chillers For Signa TwinSpeed**.

Note

Continuous water cooling is critical for the TSCC and Shield/Cryo Cooler Compressor and therefore **MUST** be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

TABLE 4-5
WATER COOLED TSCC COOLING WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TSCC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TSCC filter at cabinet inlet.
Anti-freeze	Propylene Glycol (50% maximum concentration in water)	
Inlet Pressure	150 psi maximum	
Pressure Drop	1.6 psi (11 kPa) at minimum flowrate of 4 gpm (15.14 L/min) 5.6 psi (38.5 kPa) at maximum flowrate of 20 gpm (75.70 L/min)	
Maximum Heat Output	100,000 BTU/Hr 29,307 Watts to water 12,000 BTU/Hr 3517 Watts to air	Refer also to Section 4-3 AIR COOLING REQUIREMENTS



TSCC COOLING WATER FLOWRATE/TEMPERATURE REQUIREMENT

ILLUSTRATION 4-2

4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC**Note**

Shield/Cryo Cooler Compressor Water Cooling information is provided here for the convenience of customers performing an LCC Magnet upgrade with the Signa 1.5T TwinSpeed upgrade.

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor.

The Shield/Cryo Cooler Compressor customer provided water cooling system must be closed loop design. The water cooling design can utilize open loop city water, with required filtering, only as temporary backup during loss of closed loop water cooling system. Open loop systems will not allow a chemical equilibrium to be established resulting in continual build up or etching to take place which will eventually contribute to failure. Water system capacity must be selected to insure adequate reserve for overcoming all pressure drops and still maintain the required flow rate for the Shield/Cryo Compressor.

Note

Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival, temporary water cooling must be provided if permanent site water cooling is not available.

The Shield/Cryo Cooler Compressor closed loop system may be shared with other equipment in the MR suite. The number of sharing systems should be kept to as low as possible in order to minimize contamination and reliability problems. Flow gauges and valves are recommended at all branch lines to control distribution and allow servicing of equipment.

Refer to Tables 4-3 and 4-4 for water cooling specifications.

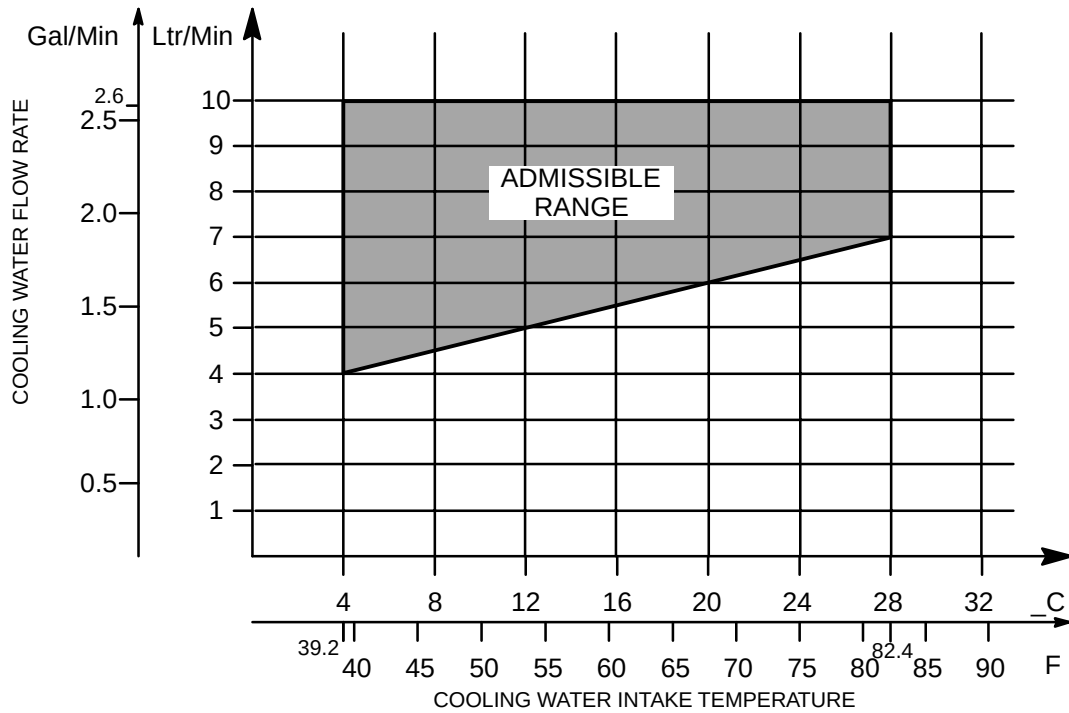
Note

The **Shield/Cryo Cooler Compressor** requires customer supplied flexible hose for mating with water cooling. Connections to the Shield/Cryo Cooler Compressor requires 0.5 in. (12.7 mm) inside diameter flexible hose and 1.0 in. (25.4 mm) adjustable compression clamps.

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4-6
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER QUALITY REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the Gradient Coil and Shield/Cryo Cooler Compressor resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install Shield/Cryo Cooler Compressor Cabinet filter at cabinet inlet.
Anti-freeze	Preferred minimum 25% by Volume Maximum 50% by volume	Use stabilized product to reduce corrosion and organic growth. Preferred minimum value to minimize organic growth.



SHIELD/CRYO COOLER COMPRESSOR COOLING WATER REQUIREMENT
ILLUSTRATION 4-3

4-4-4 . Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC (Continued)

TABLE 4-7
LCC MAGNET SHIELD/CRYO COOLER COMPRESSOR WATER COOLING SPECIFICATIONS

EQUIPMENT	INLET TEMPERATURE RANGE °F (°C)	INLET PRESSURE psi (KPa)	RECOMMENDED FLOW RATE gal/min (liter/min) See Notes 3, 4	PRESSURE DROP psi (bar) [KPa] See Note 1	TEMPERATURE RISE Δ°F (Δ°C) See Notes 1, 4, 6	TYPICAL HEAT OUTPUT BTU/Hr (Watts) See Notes 6	MAXIMUM HEAT OUTPUT BTU/Hr (Watts) See Notes 6
Shield/Cryo Cooler Compressor **	39.2 – 82.4 (4 – 28) See Note 9	Minimum 29 (200) Maximum 100 (690)	Minimum 1.1 (4) See Notes 2, 3 & 9	7.5 (0.5) [52] at minimum flow rate See Notes 2, 5, & 9	at minimum flow rate 48.4 (26.9) for 60 Hz operation 39.4 (21.9) for 50 Hz operation	25,590 (7500) for 60 Hz operation 20,728 (6100) for 50 Hz operation	28,320* (8300) for 60 Hz operation 23222* (6700) for 50 Hz operation
			Maximum 2.6 (10) See Notes 2, 3 & 9	47 (3.2) [324] at maximum flow rate See Notes 2, 5, & 9	at maximum flow rate 19.4 (10.8) for 60 Hz operation 15.7 (8.7) for 50 Hz operation		

Note: * Ensure water cooling system capacity is capable of dissipating maximum heat output.

** These water cooling specifications are the requirements **at the equipment**. The cooling system design must have allowances for pressure/temperature changes due to distance from the Shield/Cryo Cooler Compressor.

- Pressure drop and water temperature rise across equipment is given for minimum and maximum recommended flow rates as indicated. Pressure drop is measured between coolant inlet and outlet at Shield/Cryo Cooler Compressor unit.
- Water Flowmeter Kit (46-294052G1) is available to check/monitor flow rate for Shield/Cryo Cooler Compressor. Add 2 psi to total system pressure drop if flowmeter is permanently installed in system.
- Recommend a flowmeter be permanently installed in system, include flowmeter drop in total system pressure drop.
- Shield/Cryo Cooler Compressor water flow rate is based on inlet water temperature of 82.4°F (28°C), lower temperature permits lower flow. See Illustration 4-1 for graphic water temperature and flow rate admissible range.
- Minimum flow rate is for clean water (i.e. without antifreeze), maximum flow rate is for any mixture of water/antifreeze.
- Water flow rate and temperature rise value are based on water. Laboratory grade Ethylene Glycol or Propylene Glycol anti-freeze may be used (do not mix Ethylene Glycol with Propylene Glycol). Preferred concentration is 75% water and 25% Glycol to minimize organic growth. Concentration of 50/50 is acceptable with a derate of 0.8 in specific heat calculations and a 20% increase in flow.
- Pressure drop values based on new system, may rise due to calcification.
- Shield/Cryo Cooler Compressor temperature rise, typical and maximum heat output are reduced by 18% at 50 Hz operation.
- Water Cooling Circuit typical values:
 – water inlet flow 1.8 to 2.1 gal/minute (7 to 8 liter/minute)
 – water inlet temperature 53.6 to 59°F (12 to 15°C)
 There is a risk of damaging the Shield/Cryo Cooler Compressor with water inlet low temperature and low flow range.

4-4-5 Water Cooled TGWC Requirements

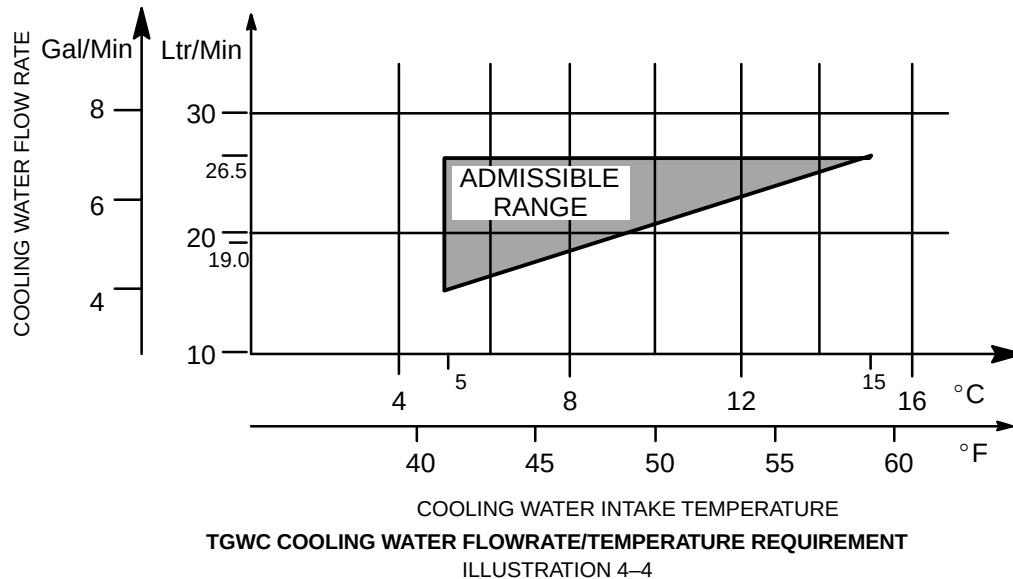
The Water Cooled configuration of the Twin Gradient Water Cooler (TGWC) contains a dedicated, closed loop, liquid-to-liquid heat exchanger system providing water cooling for the Gradient Coil ONLY. The Water Cooled TGWC configuration requires customer provided water cooling, refer to Table 4-8 for cooling water requirements. The water cooled TGWC has 0.5 inch (12.7 mm) female thread connections at the lower left side of the unit for customer water cooling connections. Refer to Illustration 2-20 for water supply and return connection location details.

Note

The TGWC provides water cooling for the Gradient Coil ONLY. Therefore customer provided water cooling is required for the water cooled Shield/Cryo Cooler Compressor. Refer to **Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC** for customer provided water cooling requirements.

TABLE 4-8
WATER COOLED TGWC WATER REQUIREMENTS

PARAMETER	REQUIREMENT	NOTES
Composition	Water	Maximum of 50% glycol in water mixture
pH level	6.5–8.2	GE recommends the use of de-ionized water to ensure longest life with fewest problems.
Hardness	less than 200 ppm of calcium carbonate	Hard water will produce calcium deposits in the TGWC resulting in decrease of cooling efficiency.
Suspended matter	less than 10 mg per liter, less than 150 micron particle size	To meet the specification for suspended matter it is necessary to install a 100–150 micron filter. Install TGWC filter at cabinet inlet.
Maximum Heat Output	37,000 BTU/Hr 10,843 Watts	Refer also to Section 4-3 AIR COOLING REQUIREMENTS
Pressure Drop	5.9 psi at minimum flowrate of 2.8 gpm 20.3 psi at maximum flowrate of 5.2 gpm	
Inlet Pressure	150 psi maximum	
Cooling Water Flowrate & Temperature	5 gpm at 41 – 59 °F (18.9 liter/min at 5 – 15 °C)	



4-5 ALTITUDE

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-5 ALTITUDE*** for additional information and specifications.

4-6 LIGHTING

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-6 LIGHTING*** for additional information and specifications.

4-7 ACOUSTICS

Note

Design improvements in the MR system (RF/Gradient Body Coil, Magnet, RF and Gradient subsystems) balances the impact of the higher performance scan protocols such that there may be slight increases in the acoustics output. The following information is supplied for design of new installations but can also be used to assess the expected performance of the upgrade MR system.

The following acoustic information is provided for site planning and architectural design activities to address acoustics to meet local regulations and customer requirements. For more information about recommended safety procedures regarding patient exposure to MR-generated acoustic levels, see the system operator manual.

4-7-1 Background

A typical MR suite has two types of acoustic noise issues. The first is the acoustics of the room where patients and technicians are impacted by the noise of the MR system as the gradients are pulsed. The second is noise transmitted to other spaces via airborne and structure borne paths.

Airborne

The airborne transmission path entails the excitation of air within the magnet room; the resonator module consisting of the magnet, RF coil, and gradient coil generates acoustic noise similar to an intense loud speaker. The airborne noise passes through walls via any openings, i.e. small holes, cracks, HVAC ducts, and waveguides, into surrounding spaces within and possibly beyond the confinements of the building. Acoustic energy can transmit across distances of significant length.

Examples of airborne acoustics issues may include the following (not limited to only these):

- MR Operator exposure at Operator Workstation
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

Structureborne

The structureborne transmission path is the result of mechanical excitation of the floor/building structure causing the building to vibrate. The vibration of the surfaces at surrounding spaces then radiates as acoustic noise. Acoustic energy can transmit across distances of significant length.

Note

Less than 5% of installed base sites have experienced structureborne acoustic issues.

Examples of structureborne acoustics issues may include the following (not limited to only these):

- Areas directly above or below the Magnet Room, may not always be an issue
- Image reading rooms adjacent to Magnet Room, may be separated by hallways
- Secretarial, offices, meeting rooms, patient rooms (ICU, exam, primary care, etc.)
- Adjacent residential areas/spaces
- In-house library facilities

4-7-2 System Acoustic Noise Levels

Any GE factory-installed protocol can be modified by operators, which can increase or decrease acoustic SPL (Sound Pressure Level); or operators may create their own protocol which could produce a higher or lower acoustic SPL as stated under **Operating Conditions** Condition 1 below. Typical scans generate acoustic levels as stated under **Operating Conditions** Condition 2 below. In addition, the exposure times are completely under operator control. Consequently, GE requires that patients wear appropriate hearing protection for all scans because levels could exceed $L_{eq} \geq 99\text{dBA}$. Again, for more information about recommended safety procedures regarding patient exposure to MR-generated acoustic noise, see the MR system operator manual.

Ambient Conditions

To reduce any background noise due to cabinet blowers, etc., acoustical ceilings, walls, and floors are recommended. The following are typical noise level readings:

- Operator Area 55 dBA
- Equipment Room 75 dBA
- Outdoor TSCC (Twin System Cooling Cabinet) 75 dBA
- Outdoor TGWC (Twin Gradient Water Cooler) 75 dBA

Operating Conditions

Condition 1

MR scanners under “worst-case” operating conditions, could generate acoustic levels (as measured at the magnet iso-center) as follows:

Average SPL 115 dBA SPL= Sound Pressure Level
 Peak 120 dB
 Frequency Range 20 to 20k Hz

Condition 2

MR scanners for many typical clinical scanning scenarios though, generate acoustic levels (as measured at the magnet iso-center) somewhat lower as follows:

Average SPL 85 to 95 dBA
 Peak 105 to 112 dB
 Frequency Range 20 to 20k Hz

As recent history has shown an evolution towards more powerful (and hence louder) gradient subsystems, architects should consider the acoustic levels stated in the “worst case” Condition 1, mentioned above. Note that high-field Signa systems have the ability to run scanning protocols which can generate acoustic levels over the entire human perceptible frequency range (20 to 20k Hz), therefore attenuation over this entire range must be considered for site design.

4-7-3 Design Guidelines

Magnet Room

Noise generated by the MR system is inherent to the operation of the system, refer to Section 4-7-2 System Acoustic Noise Levels. The sound quality (human perception) within the Magnet Room can be modified by including sound absorbing materials to make the room sound more subdued and less harsh. The measured sound levels via a sound level meter will not change. However, the measured sound levels can be reduced only when the sound level generated by the MR System is reduced.

Sound quality improvements can be achieved by the following:

- Use ceiling tiles with fiberglass panels having a 2 inch (51mm) thickness set into the standard T-bar grid system.
- Adding fiberglass panels to the side walls covering approximately 20% of the side wall surface area. The panels should focus on covering the top half of the side walls. Panels could take many different and decorative shapes to improve the sterile look of the rooms. Typically panels might be on the order of 4ft x 6ft (1.2m x 1.8m) with a thickness of 4 inches (102mm) or equivalent. Panels shape could vary to produce mosaic effects to meet the customer preference. Any decorative materials used to cover the wall panels must be porous so that sound waves can pass through with ease. In principle, a person should be able to breath through the material with ease. Fire retardant cloth should be used. The NRC (Noise Reduction Coefficient) of the panels should be 0.95 or better when mounted against a hard surface such as drywall or concrete.

Inter-Spacial Areas

Acoustic Noise Control to mitigate noise from being transmitted to other spaces often amounts to paying attention to small details while working with ordinary construction materials. The key objectives are to eliminate all cracks and gaps in the wall construction while making sure that the doors, walls, floor, and ceiling have adequate transmission loss via mass or special double wall construction along with good fitting massive doors.

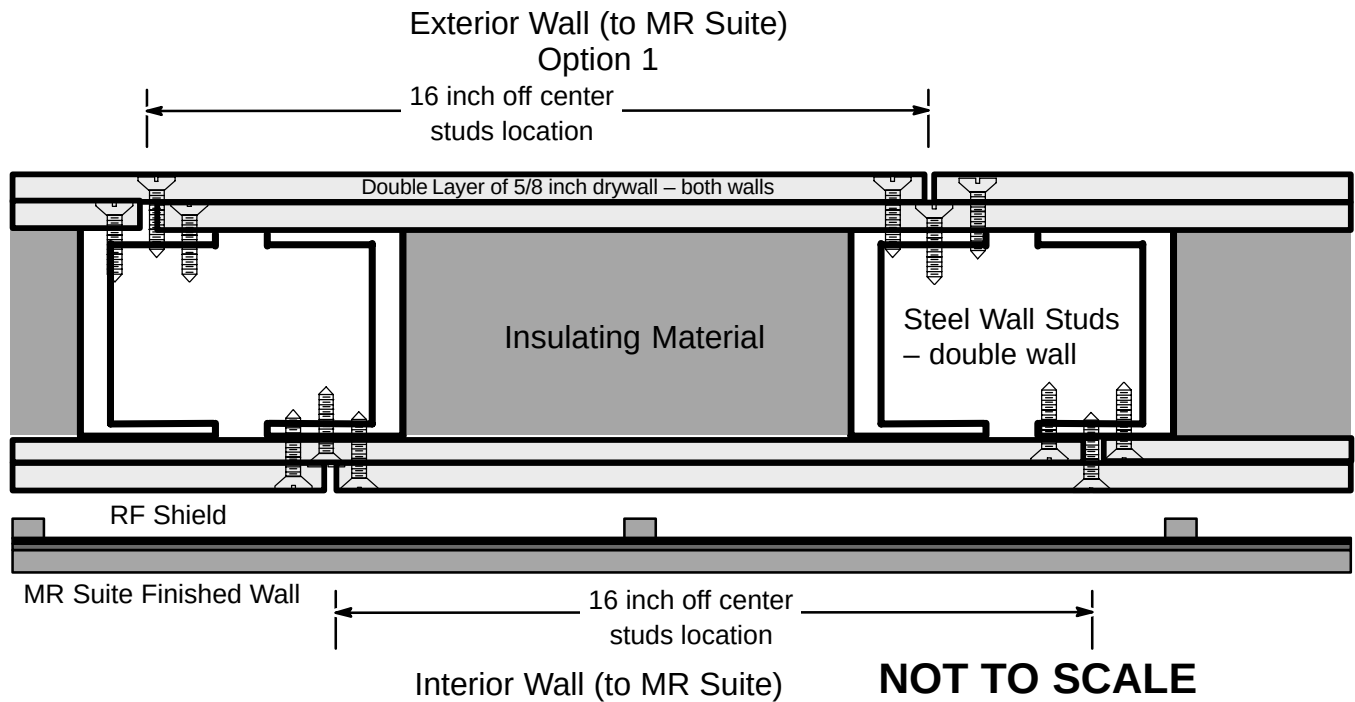
The entire Magnet must be surrounded by walls with substantial mass and/or double wall construction so that noise is contained in the room and not allowed to pass through into nearby spaces. Wall junctions must be sealed with acoustical sealant so that noise waves do not escape from the room. In principle, if the room were filled with smoke and under a positive pressure, no smoke would leak from the room.

Wall Construction

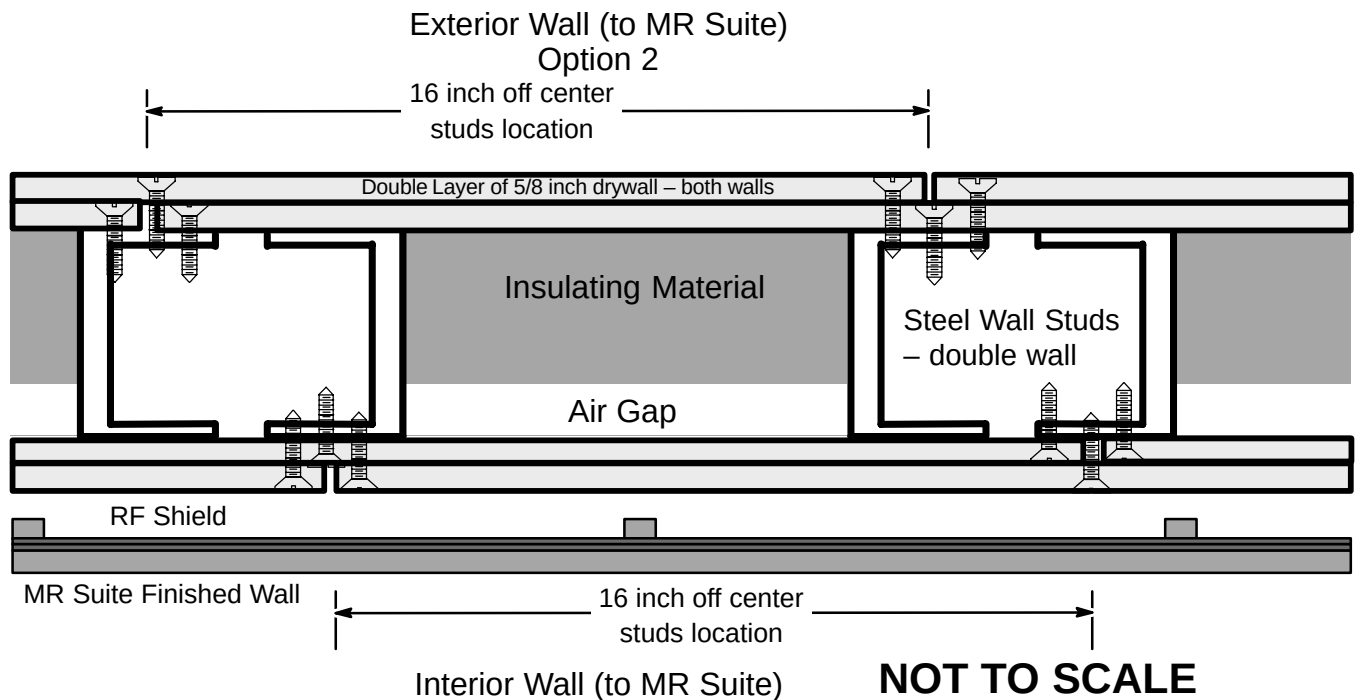
Wall Construction will entail ordinary building materials in a careful configuration.

- The preferred wall would have an ASTM STC 50 construction which entails the use of standard wall construction of steel studs (typically 3-5/8 inch (92 mm)) with 2 layers of Type X drywall (typically 5/8 (16 mm)) on each side totaling 4 layers and fiberglass batt in the stud cavity. All drywall must be overlapped by 6 inches (152 mm) or more. Beads of (USG) Acoustical Caulking (non-hardening) would be used around the entire perimeter of the drywall. Any form of wall penetration should be avoided. Any necessary wall penetrations must be sealed using combination of Acoustical Caulking (non-hardening) and fiberglass batt material. See examples of wall construction shown in Illustrations 4-5 and 4-6.
- The top of the wall must join the ceiling/floor above so that no cracks or gaps occur. If metal pan is used on the ceiling/floor (above), then flute seals would be used to seal the gaps between the drywall and the pan. Alternately drywall can be cut out to fit into the flutes. Acoustical caulking (non-hardening) will be used to seal the remaining cracks and gaps.

4-7-3 Design Guidelines (Continued)



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 1
 ILLUSTRATION 4-5



EXAMPLE OF WALL CONSTRUCTION FOR AIRBORNE NOISE CONTROL – OPTION 2
 ILLUSTRATION 4-6

4-7-3 Design Guidelines (Continued)

High Bay RF Room

A high bay RF Room is a self contained RF Room which has open air space between the RF Room ceiling and the building floor above. The air space is an acoustic transmission path. Acoustic energy must be reduced to minimize this transmission of energy through this path.

In cases where the Magnet is to be installed in a high bay, it may be most effective to enclose the RF Room with its own drywall and steel stud room. The key difference being a ceiling assembly that mimics the sidewall construction to contain noise.

- Normal high STC stud walls from above would be used to support a ceiling assembly constructed of structural C channel with two layers of drywall on each side (total of 4 layers) with fiberglass batt in the cavity.
- Penetrations should be avoided via the use of surface mounted lights. HVAC and ducts passing through the ceiling, party wall or side walls would require acoustic noise attenuation in the form of inline silencers. Gaps and cracks would be sealed between the ceiling, party wall or vertical side walls and the cryogen vent plumbing. In essence the Magnet would be enclosed in a drywall "doghouse".

Miscellaneous Plumbing, RF Windows and RF Doors

Other construction details are equally important to mitigate noise transmission to meet the intended goal.

- Pipes (gas or water) and electrical conduit or Magnet Room signal cables must be sealed where they penetrate the walls or ceiling. A heavy mastic material such as Duxseal™ is appropriate.
- RF windows should be purchased as window/frame units with an STC rating obtained from laboratory testing per ASTM standards. STC 55 or 60 windows are needed. The installation must include proper sealing to avoid sound leaks.
- RF doors should be selected to provide and STC 45 or 50 to quell the noise. Contact RF Shield Room supplier for selection of RF doors that meet the local acoustic codes and site acoustic requirements. RF door seals must be selected to prevent small gaps around the door perimeter and at the door threshold. RF door seals would either require periodic replacement or a door seal that would last the life of the Magnet Room.

4-8 ROOM VENTILATION

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-8 ROOM VENTILATION*** for requirements and recommendations.

4-9 CRYOGENIC VENTING



WARNING!

IF AN UPGRADE TO AN LCC MAGNET IS BEING PERFORMED WITH THE SIGNA TWINSPEED SYSTEM UPGRADE THEN REFER TO ***DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION SECTION 4-9 CRYOGENIC VENTING FOR REQUIREMENTS AND RECOMMENDATIONS FOR LCC MAGNET CRYOGENIC VENTING.***

4-10 ALARM DEVICES, WATER SENSORS AND THERMOSTATS

4-10-1 System Cabinet

For Signa TwinSpeed with EXCITE Technology the System Cabinet has 104°F (40°C) and 122°F (50°C) temperature sensors. When the temperature in the System Cabinet reaches 104°F (40°C) a Warning Message will be displayed on the Operator Workspace. When the temperature in the System Cabinet reaches 122°F (50°C) or above a Warning Message will be displayed on the Operator Workspace and the system scanning operation will be inhibited. Normal scanning operation will be allowed when the System Cabinet temperature returns to normal. Any external alarm device for room temperature monitoring must be supplied by the customer.

For Signa TwinSpeed the System Cabinet has one temperature sensor and one control module which sounds an alarm located in the System Cabinet when temperature reaches 94°F (34.4°C) and a Warning Message will be displayed on the Operator Workspace. After 3 minutes of alarm condition PDU will revert to full off condition. Any external alarm device other than mentioned above must be supplied by the customer.

4-11 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS.***

4-12 CONSTRUCTION MATERIALS

If an upgrade to an LCC magnet is being performed with the Signa TwinSpeed system upgrade then refer to ***Direction 2294003 Signa TwinSpeed Pre-Installation Section 4-14 CONSTRUCTION MATERIALS*** for requirements and recommendations.

4-13 VIBRATION

4-13-1 Definition

Improvements in MR imaging technology has increased imaging capabilities. Certain MR procedures require a stable environment to achieve high resolution image quality. **Environmental site vibration has the ability to affect the magnet phase stability and ultimately image quality.** The vibration effects on image quality can be minimized early in the site planning for Signa Horizon upgrades.

The magnet may be sensitive to vibration in the frequency range of 0.5 to 45 Hz depending on the amplitude of the vibration. In the physical area where the MR system is located, every precaution must be taken to ensure that the vibration is minimized. In the magnet siting area, the structural stability and behavioral characteristics can be assessed. **The stability tests (HSS) currently used by local GE Field Service to evaluate the existing system performance and image quality can be used to assess the vibration environment. Sites which currently pass their system stability criteria (HSS) may proceed with the upgrade. Sites which have marginal stability performance require vibration measurements to be performed to identify vibration source(s).** Then it is the customer's responsibility to contract a vibration consultant to perform vibration measurements as defined in Appendix A, MR SITE VIBRATION TEST GUIDELINES. With the vibration consultant, local GE Field Service and/or Installation Specialist must verify the elimination/reduction of all identified sources do improve the vibration environment. GE can assist in interpreting marginal site test results and predicting the impact on system performance. However it is ultimately the customer/architect/engineer responsibility to design a site solution.

**WARNING!**

REFER TO *DIRECTION 2294003 SIGNA TWINSPEED PRE-INSTALLATION* SECTION 4-15 VIBRATION SPECIFICATIONS FOR SIGNA 1.5T TWINSPEED SYSTEMS.